

1. (a) Find the exact value of the following expression

$$\cosh[\tanh^{-1}(\frac{1}{2})] \times \cosh[\tanh^{-1}(\frac{1}{3})] - \sinh[\tanh^{-1}(\frac{1}{2})] \times \sinh[\tanh^{-1}(\frac{1}{3})]$$

(b) If $f(x) = \sqrt[4]{\frac{1+\tanh x}{1-\tanh x}} - \sinh(\frac{x}{4}) \cosh(\frac{x}{4})$, find $f'(\ln 9)$

(sol)

(a) Since $\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}$, $|x| < 1$,

$$\tanh^{-1}(\frac{1}{2}) = \frac{1}{2} \ln \frac{1+\frac{1}{2}}{1-\frac{1}{2}} = \ln \sqrt{3}, \text{ and } \tanh^{-1}(\frac{1}{3}) = \frac{1}{2} \ln \frac{1+\frac{1}{3}}{1-\frac{1}{3}} = \ln \sqrt{2}.$$

$$\therefore \cosh[\tanh^{-1}(\frac{1}{2})] \times \cosh[\tanh^{-1}(\frac{1}{3})] - \sinh[\tanh^{-1}(\frac{1}{2})] \times \sinh[\tanh^{-1}(\frac{1}{3})]$$

$$= \cosh[\tanh^{-1}(\frac{1}{2}) - \tanh^{-1}(\frac{1}{3})]$$

$$= \cosh(\ln \sqrt{3} - \ln \sqrt{2})$$

$$= \cosh\left(\ln \sqrt{\frac{3}{2}}\right)$$

$$= \frac{e^{\ln(\frac{\sqrt{3}}{\sqrt{2}})} + e^{-\ln(\frac{\sqrt{3}}{\sqrt{2}})}}{2}$$

$$= \frac{\frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{2}}{\sqrt{3}}}{2} = \frac{5}{2\sqrt{6}}$$

$$\begin{aligned} (b) \sqrt[4]{\frac{1+\tanh x}{1-\tanh x}} &= \sqrt[4]{\frac{1+\frac{\sinh x}{\cosh x}}{1-\frac{\sinh x}{\cosh x}}} \\ &= \sqrt[4]{\frac{\cosh x + \sinh x}{\cosh x - \sinh x}} \\ &= \sqrt[4]{\frac{e^x}{e^{-x}}} \\ &= e^{\frac{1}{2}x} \end{aligned}$$

and

$$\sinh(\frac{x}{4}) \cosh(\frac{x}{4}) = \frac{1}{2} \sinh(\frac{x}{2})$$

$$\therefore f(x) = e^{\frac{1}{2}x} - \frac{1}{2} \sinh(\frac{x}{2})$$

$$\Rightarrow f'(x) = \frac{1}{2} e^{\frac{1}{2}x} - \frac{1}{4} \cosh(\frac{x}{2})$$

$$\therefore f'(\ln 9) = \frac{1}{2} e^{\frac{1}{2} \ln 9} - \frac{1}{4} \cosh(\frac{\ln 9}{2}) = \frac{13}{12}$$

2. (a) Find $\lim_{x \rightarrow \infty} (\ln x)^{\frac{\ln x}{x}}$

(b) Find linearization $h(x)$ of $f(x) = \sec x$ at $a = 0$ and use it to estimate $\sec\left(\frac{1}{\sqrt{2019}}\right)$

(sol)

(a) let $y = (\ln x)^{\frac{\ln x}{x}}$

$$\Rightarrow \ln y = \frac{\ln x}{x} \ln(\ln x) = \frac{(\ln x) \ln(\ln x)}{x}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{(\ln x) \ln(\ln x)}{x}$$

$$\stackrel{\text{L'Hosp}}{=} \lim_{x \rightarrow \infty} \left\{ \frac{1}{x} \ln(\ln x) + \cancel{\ln x} \cdot \frac{1}{\cancel{\ln x}} \cdot \frac{1}{x} \right\}$$

$$= \lim_{x \rightarrow \infty} \frac{\ln(\ln x) - 1}{x}$$

$$\stackrel{\text{L'Hosp}}{=} \lim_{x \rightarrow \infty} \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$= 0$$

$$\therefore \lim_{x \rightarrow \infty} (\ln x)^{\frac{\ln x}{x}} = \lim_{x \rightarrow \infty} y = e^0 = 1$$

(b) $f(x) = \sec x \Rightarrow f'(x) = \sec x \tan x$

$$\Rightarrow f(0) = 1 \text{ and } f'(0) = 0$$

$\therefore$ The linearization $h(x)$ of $f(x)$ at $a = 0$ is

$$h(x) = f(0) + f'(0)(x - 0)$$

$$= 1$$

and the linear approximation of $f(x)$ at $a = 0$ is

$$\sec x \approx h(x) = 1$$

$$\therefore \sec\left(\frac{1}{\sqrt{2019}}\right) \approx 1$$

$$3. \text{ Find } \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x^3}} \int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt$$

$$(\text{sol}) \quad \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x^3}} \int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt$$

$$= \lim_{x \rightarrow 0^+} \frac{\int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2) dt}{\sqrt{x^3}}$$

$$\stackrel{\frac{0}{0}}{\underset{\text{L'Hosp.}}{=}} \lim_{x \rightarrow 0^+} \frac{\frac{1}{\sqrt{x}} \sin(4x) - \frac{1}{2\sqrt{x}} \sin x}{\frac{3}{2} \sqrt{x}} \quad (\text{by FTC 1})$$

$$= \frac{1}{3} \lim_{x \rightarrow 0^+} \frac{2 \sin 4x - \sin x}{3x}$$

$$\stackrel{\frac{0}{0}}{\underset{\text{L'Hosp.}}{=}} \frac{1}{3} \lim_{x \rightarrow 0^+} \frac{8 \cos 4x - \cos x}{3}$$

$$= \frac{7}{3}$$

4. The solution between 3 and 4 of the equation $\tan x = 0$ is π . Starting with $x_1 = 3$, find the second approximation x_2 to π (Use $\rho' 6 = -0.30$)

(sol) Let $f(x) = \tan x$

$$\Rightarrow f'(x) = \sec^2 x$$

$$\begin{aligned}\therefore x_{n+1} &= x_n - \frac{\tan x_n}{\sec^2 x_n} \\ &= x_n - \frac{\frac{\rho' x_n}{\omega x_n}}{\frac{1}{\omega^2 x_n}} \\ &= x_n - \rho' x_n \omega x_n \\ &= x_n - \frac{1}{2} \rho' 2 x_n\end{aligned}$$

$$\Rightarrow x_1 = 3$$

$$\begin{aligned}x_2 &= x_1 - \frac{1}{2} \rho' 2 x_1 \\ &= 3 - \frac{1}{2} \rho' 6 \\ &= 3 - \frac{1}{2} (-0.30) \\ &= 3.15\end{aligned}$$

5. let $I_n = \int_0^1 (1-x^2)^n dx$ where n is a positive integer.

(a) Find the number k such that $I_{20} = k I_{19}$

(b) Find I_{20}

(sol)

$$(a) I_n = \int_0^1 (1-x^2)^n dx$$

$$= \left[x (1-x^2)^n \right]_0^1 - \int_0^1 x \cdot n (1-x^2)^{n-1} (-2x) dx$$

$$= 2n \int_0^1 x^2 (1-x^2)^{n-1} dx$$

$$= 2n \int_0^1 [1 - (1-x^2)] (1-x^2)^{n-1} dx$$

$$= 2n \int_0^1 (1-x^2)^{n-1} dx - 2n \int_0^1 (1-x^2)^n dx$$

$$= 2n I_{n-1} - 2n I_n$$

$$\Rightarrow (2n+1) I_n = 2n I_{n-1}$$

$$\Rightarrow I_n = \frac{2n}{2n+1} I_{n-1}$$

$$\therefore I_{20} = \frac{40}{41} I_{19}$$

$$\Rightarrow k = \frac{40}{41}$$

$$(b) I_{20} = \frac{40}{41} I_{19}$$

$$= \frac{40}{41} \cdot \frac{38}{39} I_{18}$$

$$= \dots$$

$$= \frac{40}{41} \cdot \frac{38}{39} \cdot \dots \cdot \frac{4}{5} \cdot \frac{2}{3} I_0$$

$$= \frac{(40 \cdot 38 \cdot \dots \cdot 4 \cdot 2)^2}{41!} \quad \left(\because I_0 = \int_0^1 (1-x^2)^0 dx = 1 \right)$$

$$= \frac{2^{40} \cdot (20!)^2}{41!}$$

6. Suppose the graph of the function $y = x^3 + px^2 + qx$ intersects the parabola $y = x^2$ when $x = 0$, $x = 1$ and $x = a$, where $a > 1$. If the two regions between the curves have the same area, find the volume of the solid obtained by rotating the two regions about the y -axis.

(sol) The equation $x^3 + px^2 + qx = x^2$ has solutions $x = 0, 1, a$, that is, $x^3 + (p-1)x^2 + qx = x(x-1)(x-a)$

Since the two region between the curves have the same area,

$$\int_0^a \{x^3 + (p-1)x^2 + qx\} dx = 0$$

$$\Rightarrow a = 2, p = -2, q = 2$$

$\therefore$ The volume V is

$$\begin{aligned} V &= \int_0^1 2\pi x (x^3 - 3x^2 + 2x) dx + \int_1^2 2\pi x (-x^3 + 3x^2 - 2x) dx \\ &= 2\pi \left[\int_0^1 (x^4 - 3x^3 + 2x^2) dx + \int_1^2 (-x^4 + 3x^3 - 2x^2) dx \right] \\ &= 2\pi \left[\left[\frac{1}{5}x^5 - \frac{3}{4}x^4 + \frac{2}{3}x^3 \right]_0^1 + \left[-\frac{1}{5}x^5 + \frac{3}{4}x^4 - \frac{2}{3}x^3 \right]_1^2 \right] \\ &= \pi. \end{aligned}$$

7. Evaluate the indefinite integral:

$$\int \frac{1}{x \sqrt{3x^2 - 2x - 1}} dx \quad (\text{Hint: Substitute } x = \frac{1}{t})$$

(sol) Let $x = \frac{1}{t}$

$$\Rightarrow dx = -\frac{1}{t^2} dt$$

$$\therefore \int \frac{dx}{x \sqrt{3x^2 - 2x - 1}} = \int \frac{1}{\frac{1}{t} \sqrt{\frac{3}{t^2} - \frac{2}{t} - 1}} \left(-\frac{1}{t^2}\right) dt$$

$$= - \int \frac{1}{\sqrt{3 - 2t - t^2}} dt$$

$$= - \int \frac{1}{\sqrt{2^2 - (t+1)^2}} dt$$

$$\left(\begin{array}{l} t+1 = 2 \sin \theta \\ dt = 2 \cos \theta d\theta \end{array} \right)$$

$$= - \int \frac{1}{\sqrt{2^2 - 2^2 \sin^2 \theta}} 2 \cos \theta d\theta$$

$$= - \int d\theta$$

$$= -\theta + C$$

$$= -\sin^{-1} \frac{t+1}{2} + C$$

$$= -\sin^{-1} \frac{x+1}{2x} + C$$

8. Evaluate the definite integral:

$$\int_0^{\pi} \left[\frac{\omega \theta x}{\omega \theta x + \omega \theta (\pi - x)} + \frac{x \rho' x}{1 + \omega^2 x} \right] dx$$

(sol) i) Let $u = \pi - x \Rightarrow \begin{pmatrix} du = -dx (\Rightarrow dx = -du) \\ x = 0 \rightarrow u = \pi \\ x = \pi \rightarrow u = 0 \end{pmatrix}$

$$\begin{aligned} I &= \int_0^{\pi} \frac{\omega \theta x}{\omega \theta x + \omega \theta (\pi - x)} dx = \int_{\pi}^0 \frac{\omega \theta (\pi - u)}{\omega \theta (\pi - u) + \omega \theta u} (-du) \\ &= \int_0^{\pi} \frac{\omega \theta (\pi - u)}{\omega \theta (\pi - u) + \omega \theta u} du \\ &= \int_0^{\pi} \frac{\omega \theta (\pi - x)}{\omega \theta x + \omega \theta (\pi - x)} dx \end{aligned}$$

$$\begin{aligned} \therefore 2I &= \int_0^{\pi} \frac{\omega \theta x + \omega \theta (\pi - x)}{\omega \theta x + \omega \theta (\pi - x)} dx \\ &= \pi \end{aligned}$$

$$\Rightarrow \int_0^{\pi} \frac{\omega \theta x}{\omega \theta x + \omega \theta (\pi - x)} dx = \frac{\pi}{2}$$

(ii) Let $x = \frac{\pi}{2} - u \Rightarrow \begin{pmatrix} dx = -du \\ x = 0 \rightarrow u = \frac{\pi}{2} \\ x = \pi \rightarrow u = -\frac{\pi}{2} \end{pmatrix} \quad \begin{aligned} \omega x &= \omega \left(\frac{\pi}{2} - u \right) = \rho' u \\ \rho' x &= \rho' \left(\frac{\pi}{2} - u \right) = \omega u \end{aligned}$

$$\begin{aligned} \therefore \int_0^{\pi} \frac{x \rho' x}{1 + \omega^2 x} dx &= \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - u \right) \omega u}{1 + \rho'^2 u} (-du) \\ &= \frac{\pi}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\omega u}{1 + \rho'^2 u} du - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{u \omega u}{1 + \rho'^2 u} du \\ &= \frac{\pi}{2} \cdot 2 \int_0^{\frac{\pi}{2}} \frac{1}{1 + \rho'^2 u} d(\rho' u) \\ &= \pi \left[\tan^{-1}(\rho' u) \right]_0^{\frac{\pi}{2}} \\ &= \pi (\tan^{-1} 1 - \tan^{-1} 0) \\ &= \frac{\pi^2}{4} \end{aligned}$$

$$\therefore \int_0^{\pi} \left[\frac{\omega \theta x}{\omega \theta x + \omega \theta (\pi - x)} + \frac{x \rho' x}{1 + \omega^2 x} \right] dx = \frac{\pi}{2} + \frac{\pi^2}{4}$$

9. Evaluate the indefinite integral:

$$\int \left(\frac{1}{\sec^2 x \cos^2 x} + \frac{\sec^2 x}{\tan^3 x - 1} \right) dx$$

$$\begin{aligned} \text{(sol) (i)} \quad \int \frac{1}{\sec^2 x \cos^2 x} dx &= \int \frac{\sec^2 x + \cos^2 x}{\sec^2 x \cos^2 x} dx \\ &= \int (\sec^2 x + \csc^2 x) dx \\ &= \tan x - \cot x + C \end{aligned}$$

$$\begin{aligned} \text{(ii) Let } u &= \tan x \\ \Rightarrow du &= \sec^2 x dx \end{aligned}$$

$$\therefore \int \frac{\sec^2 x}{\tan^3 x - 1} dx = \int \frac{1}{u^3 - 1} du$$

$$\begin{aligned} \left(\begin{aligned} \frac{1}{u^3 - 1} &= \frac{1}{(u-1)(u^2+u+1)} \\ &= \frac{A}{u-1} + \frac{Bu+C}{u^2+u+1} \\ \Rightarrow 1 &= A(u^2+u+1) + (Bu+C)(u-1) \\ \Rightarrow A &= \frac{1}{3}, B = -\frac{1}{3}, C = -\frac{2}{3} \end{aligned} \right. \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3} \int \left(\frac{1}{u-1} - \frac{u+2}{u^2+u+1} \right) du \\ &= \frac{1}{3} \int \frac{1}{u-1} - \frac{\frac{1}{2}(2u+1) + \frac{3}{2}}{u^2+u+1} du \\ &= \frac{1}{3} \int \left(\frac{1}{u-1} - \frac{1}{2} \frac{2u+1}{u^2+u+1} - \frac{3}{2} \frac{1}{(u+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \right) du \\ &= \frac{1}{3} \ln |u-1| - \frac{1}{6} \ln |u^2+u+1| \\ &\quad - \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2}{\sqrt{3}} \left(u + \frac{1}{2} \right) \right) + C \\ &= \frac{1}{3} \ln |\tan x - 1| - \frac{1}{6} \ln |\tan^2 x + \tan x + 1| \\ &\quad - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} (2 \tan x + 1) \right) + C \end{aligned}$$